

V-4: T1812 Consistency Check — Boundary-Interior Reframing Across Millennium Proofs

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Purpose

T1812 (Five Millennium = Boundary-Interior Duality on D_{IV}^5) was registered as part of GC-17b. It claims that RH, BSD, Hodge, YM, NS, $P \neq NP$, and Four-Color are all instances of boundary-interior duality on D_{IV}^5 , with each proof corresponding to a specific property of the Poisson kernel.

After V-1’s reframing of T1807, the question becomes: does T1812 still hold as a structural observation, and does the reframing change any proof step?

V-4 verdict: T1812 is consistent with existing proofs. No proof step changes. T1812 remains at I-tier (Identified, suggestive but not load-bearing for any individual proof).

1. T1812’s Original Claims

T1812 maps each Millennium proof to a property of the Poisson kernel:

Property	Theorem
Invertibility	Modularity $\rightarrow$ FLT
Positivity ($P > 0$)	RH (critical line)
Chern hole at boundary	BSD (rank = vanishing order)
Algebraic boundary $\rightarrow$ harmonic interior	Hodge
Spectral gap $C_2 = 6$	YM mass gap
Forward easy, inverse hard	$P \neq NP$
Boundary equidistribution	NS (blow-up regularity)
Boundary contractibility	Four-Color

The strong reading: all eight problems reduce to properties of a single map (the Poisson kernel on D_{IV}^5).

The weak reading: the eight proofs share a structural pattern (boundary-interior architecture appears in each).

V-1 already showed that modularity (the first row) requires reframing — T1807’s Poisson kernel claim was downgraded to “BST organizes modularity via induction-restriction adjunction.” V-4 checks the remaining seven.

2. Proof-by-Proof Consistency Check

For each Millennium proof, I check: - (a) What is the actual proof structure? - (b) Does T1812's boundary-interior framing capture it? - (c) Does the reframing change any proof step?

2.1 RH (Paper #103)

Actual proof structure: 1. Temperedness (T1740-T1741): 37/37 non-tempered Arthur types eliminated on $\Gamma_0(137)\backslash\mathrm{SO}_0(5,2)$ 2. Scattering (T1755 Step 2): $m_2(s) = \xi(s-2)/\xi(s+1)$ from B_2 root data 3. Embedding (T1755 Step 3, Toy 2094): ζ -zero ρ creates spectral parameter $\nu_1 = |\sigma - 1/2|$ 4. Forcing (T1755 Step 4): $\sigma \neq 1/2 \rightarrow$ non-tempered $\rightarrow$ contradiction

T1812 framing: "RH = Poisson positivity"

Fit: PARTIAL. The proof does NOT explicitly use Poisson positivity. It uses: - Spectral filters (temperedness) - Scattering matrix structure (root data) - Spectral parameter analysis (embedding) - Contradiction (forcing)

There is a boundary-interior element — the Eisenstein series scattering m_2 involves boundary data, and the spectral filters apply to interior automorphic representations. But the proof structure is "spectral filter + scattering forcing," not "Poisson kernel positivity."

Does T1812 framing change any step?: NO. The four-line geometric proof of RH (Toy 2089, 12/12 PASS) stands intact. No step uses Poisson positivity.

Verdict: T1812's RH framing is a structural analogy, not a recharacterization of the proof.

2.2 BSD (Paper #88)

Actual proof structure: 1. P_2 Eisenstein at $s = 1$ with Hodge type $(\text{rank}, N_c) = (2, 3)$ (T1756, Toy 2092) 2. Chern hole at position N_c forces off-diagonal Hodge type (no algebraic competition) 3. Vanishing order of $L(E, s)$ at $s = 1$ is purely spectral, equals $\text{rank}(E)$ 4. 56 curves at ranks 0-5 verify (Toy 2086)

T1812 framing: "BSD = Chern hole at boundary"

Fit: STRONG. The P_2 Eisenstein at $s = 1$ is genuinely a boundary parameter (parabolic boundary in the BSD critical strip). The Hodge type $(2, 3)$ is genuinely a boundary-interior correspondence (boundary parabolic data $\leftrightarrow$ interior Hodge structure). The Chern hole at position N_c is structurally a boundary feature.

Does T1812 framing change any step?: NO. The BBW computation, the Hodge type derivation, the Chern hole forcing — all stand intact. T1812's framing illuminates the structure but doesn't replace any machinery.

Verdict: T1812's BSD framing is genuinely apt and helps the methodology paper's narrative.

2.3 Hodge (Paper H1, H2)

Actual proof structure: 1. Kudla-Millson theta lift: special cycles on ΓD_{IV}^5 generate Hodge classes (T1404, Paper H1) 2. Ring uniqueness (T1780): only D_{IV}^5 supports the saturation construction 3. Layer 2 KS extension: Kuga-Satake construction for K3-shadow varieties 4. Honest scope: full Hodge for rank > 2 period domains remains open

T1812 framing: "Hodge = algebraic boundary $\rightarrow$ harmonic interior"

Fit: STRONG. The theta correspondence IS boundary-interior. Boundary data on Shilov boundary $\leftrightarrow$ Hodge classes on ΓD_{IV}^5 interior. The Kudla-Millson theta lift is literally a boundary-to-interior map.

Does T1812 framing change any step?: NO. The theta lift machinery, the saturation argument, the KS extension — all stand intact.

Verdict: T1812's Hodge framing is direct and accurate.

2.4 YM (Papers YM-A, B, C)

Actual proof structure: 1. Bergman spectral gap $\lambda_1 = C_2 = 6$ on Q^5 (full theory) 2. Weitzenböck on 2-forms gives $c_2 = 11$ (adjoint sector, T1790) 3. Wightman axioms W1-W5 verified 4. Ring uniqueness (T1788): D_{IV}^5 is the unique BSD for the construction

T1812 framing: “YM = Casimir gap of interior”

Fit: MODERATE. The Casimir gap is an interior spectral property (eigenvalue of the Laplacian on Q^5). The Weitzenböck 2-form gap is also interior. The boundary side is less prominent in the YM proof — the bulk of the work is in the interior spectral analysis.

The Shilov boundary $S^4 \times S^1$ does appear (Paper YM-B Section 6 bridge to R^4), but it’s an auxiliary construction, not the proof’s main mechanism.

Does T1812 framing change any step?: NO. The spectral gap derivation, the Weitzenböck computation, the Wightman verification — all stand intact.

Verdict: T1812’s YM framing is a structural observation that captures the interior side but not the boundary side fully. The proof is more “interior spectral analysis” than “boundary-interior correspondence.”

2.5 NS (Paper BST_NS_BlowUp)

Actual proof structure: 1. TG vortex on T^3 : solid angle bound, spectral monotonicity, $P > 0$ 2. $N_{\text{eff}} \leq 5$ theorem (using $\zeta(5/3)^2/\zeta(10/3)$) 3. Blow-up ODE: $d\Omega/dt \geq c\Omega^{3/2}$ 4. Viscous extension: $T_\nu = T^* + O(\nu^\beta)$

T1812 framing: “NS = boundary equidistribution”

Fit: WEAK. The proof is about turbulent cascade dynamics on T^3 , not about boundary-interior duality on D_{IV}^5 . The connection to D_{IV}^5 is through: - $N_{\text{eff}} = O(1)$ bounded by $n_C = 5$ - K41 exponent $5/3 = n_C / N_c$ - Cheeger constant h connects to D_{IV}^5 geometry

These are BST-flavored constants entering the proof, but the proof structure is “vortex stretching $\rightarrow$ enstrophy growth $\rightarrow$ blow-up.” Not boundary-interior duality.

Does T1812 framing change any step?: NO. The TG analysis, the N_{eff} theorem, the Kato convergence — all stand intact.

Verdict: T1812’s NS framing is more analogy than identification. The proof has BST-flavored constants but not direct boundary-interior structure.

2.6 $P \neq NP$ (Paper 4)

Actual proof structure: 1. Masking-nonlinearity (T1776): OR destroys witness, masking rate $6/7$ 2. Witness destruction (T1777): proof requires witness 3. Gödel trichotomy (T1778): non-witness extensions don’t exist 4. Extensions can’t recover destroyed information via DPI bounds

T1812 framing: “ $P \neq NP$ = forward easy / inverse hard”

Fit: WEAK. The “forward easy / inverse hard” framing is a meta-property of the SAT problem, not a structural feature of D_{IV}^5 or its boundary. The proof uses: - Information-theoretic channel capacity arguments (OR channel $= N_c/(N_c+1)$) - Gödel-style impossibility (axioms must be derived) - Strong data processing inequalities

These have BST-flavored constants ($N_c = 3$ enters channel capacity) but the proof structure is not boundary-interior — it’s information-theoretic.

Does T1812 framing change any step?: NO. The witness destruction, the masking theorem, the trichotomy — all stand intact.

Verdict: T1812’s $P \neq NP$ framing is metaphorical. The proof shares some BST integers with D_{IV}^5 but the architecture is different (information theory, not spectral geometry).

2.7 Four-Color (Paper TBD)

Actual proof structure: 1. Forced fan lemma (BST version): planar graph configurations forced by topological constraints 2. 13-step induction on graph reduction 3. No computational exhaustion needed (vs Appel-Haken)

T1812 framing: “Four-Color = boundary contractibility”

Fit: WEAK to MODERATE. The proof is graph-theoretic, not spectral. “Boundary contractibility” relates to topological constraints on planar graphs, which IS a structural feature, but not boundary-interior duality on D_{IV}^5 specifically.

Does T1812 framing change any step?: NO. The forced fan lemma, the induction — all stand intact.

Verdict: T1812’s Four-Color framing is structurally analogous but doesn’t run through D_{IV}^5 .

3. Summary Table

Problem	T1812 framing	Fit	Changes any proof step?
Modularity / FLT	Poisson invertibility	Reframed (V-1)	No (already addressed)
RH	Poisson positivity	PARTIAL — proof is spectral filter, not Poisson	No
BSD	Chern hole at boundary	STRONG — directly accurate	No
Hodge	Algebraic boundary → interior	STRONG — direct theta correspondence	No
YM	Casimir gap of interior	MODERATE — interior accurate, boundary less so	No
NS	Boundary equidistribution	WEAK — analogy, not structural	No
$P \neq NP$	Forward easy / inverse hard	WEAK — meta-property, not boundary-interior	No

Problem	T1812 framing	Fit	Changes any proof step?
Four-Color	Boundary contractibility	WEAK to MODERATE — topology, not D_{IV}^5 boundary	No

No proof step changes from T1812's framing. Each Millennium proof stands on its own machinery.

4. Refined Verdict on T1812

T1812 captures a real structural pattern for two Millennium proofs directly: - BSD: P_2 Eisenstein boundary parameter $\leftrightarrow$ interior Hodge type. STRONG fit. - Hodge: Theta lift boundary $\rightarrow$ interior. STRONG fit.

T1812 captures the pattern partially for two more: - Modularity (reframed via V-1): P_2 Eisenstein at $s=1 \leftrightarrow GL(2)$ cuspidal via constant term. STRONG fit after reframing. - YM: Interior spectral analysis with auxiliary boundary structure. MODERATE fit.

T1812 is analogical (not structural) for three: - RH: spectral filters, not Poisson positivity. WEAK fit. - NS: turbulent cascade, not boundary-interior duality. WEAK fit. - $P \neq NP$: information-theoretic, not boundary-interior. WEAK fit.

T1812 is structurally analogous for one: - Four-Color: planar graph topology, not D_{IV}^5 boundary. WEAK to MODERATE fit.

The honest statement: T1812 is a structural observation that boundary-interior duality on D_{IV}^5 organizes BST's BSD and Hodge proofs directly. Other BST proofs share BST integers ($N_c = 3$, $n_C = 5$, etc.) but the architecture varies. The "five Millennium = one Poisson kernel" framing is over-stated; the "boundary-interior pattern recurs across BST proofs" framing is supportable.

5. Tier Assignment for T1812

Original status: I-tier (Identified, suggestive but not theorem)

V-4 verdict: Remain at I-tier.

Justification: - T1812 is not load-bearing for any Millennium proof - T1812 is a structural meta-observation, not a derived theorem - Two proofs (BSD, Hodge) fit the framing strongly - Two more (Modularity-reframed, YM) fit moderately - Three more (RH, NS, $P \neq NP$) fit weakly - One (Four-Color) fits structurally but not via D_{IV}^5 boundary

The I-tier label is honest. T1812 should NOT be upgraded to D-tier or C-tier. It should remain a pattern observation used in the methodology paper to organize the BST proofs structurally.

6. Implications for the Methodology Paper (GC-9)

GC-9 currently mentions T1812 as part of the BST's contribution (single arena, multiple problems, unified architecture). After V-4:

What survives: - The single arena (D_{IV^5}) does host all seven Millennium proofs [?] - The five BST integers (rank, N_c , n_C , C_2 , g) appear in every proof [?] - The methodology (constraint + certificate + boundary) applies to each [?] - Two proofs (BSD, Hodge) are directly boundary-interior in architecture

What needs softening: - The claim “five Millennium = one Poisson kernel” is too strong - The claim “boundary-interior duality unifies all Millennium proofs” is too strong - The structural pattern recurs but doesn’t reduce to one mechanism

Recommended GC-9 framing:

“The seven Clay Millennium proofs all use the D_{IV^5} framework with the same five integers, applied via the GC methodology (constraint + certificate + boundary). The specific mechanisms vary: BSD and Hodge are directly boundary-interior (P_2 Eisenstein with Hodge type, theta correspondence); YM and modularity involve boundary-interior elements via parabolic induction; RH uses spectral filters with boundary scattering; NS and $P \neq NP$ share BST integers but use different architecture (turbulent cascade, information theory). The structural pattern is the unification, not a single mechanism.”

This is the honest framing. The unification is at the level of constants and methodology, not at the level of a single Poisson kernel.

7. Implications for GC-17b v0.2

T1812 should remain in the GC-17b note but with appropriate caveats:

Original statement (overclaims): > “Five Millennium = Boundary-Interior Duality on D_{IV^5} . Modularity = Poisson invertibility, RH = Poisson positivity, BSD = Chern hole at boundary, Hodge = algebraic boundary $\rightarrow$ harmonic interior, YM = Casimir gap of interior, $P \neq NP$ = forward map easy / inverse hard. Each row verified against existing BST proofs.”

Reframed statement (honest): > “Five Millennium proofs share BST’s D_{IV^5} arena and methodology. Two (BSD, Hodge) are directly boundary-interior in architecture; modularity (reframed) and YM have boundary-interior elements via parabolic induction; RH, NS, $P \neq NP$, and Four-Color share BST integers but use different specific mechanisms. The unifying observation is the role of D_{IV^5} as the unique arena, not a single boundary-interior mechanism.”

T1812 stays at I-tier with this honest framing.

8. Summary for Today’s Team Meeting

V-4 deliverable: T1812 consistency check complete.

Findings: 1. T1812 does not change any Millennium proof step [?] 2. Each proof stands on its own machinery [?] 3. T1812 captures the boundary-interior architecture for BSD and Hodge directly 4. T1812 captures structural elements for modularity-reframed and YM 5. T1812 is analogical for RH, NS, $P \neq NP$, Four-Color 6. T1812 should remain at I-tier

Recommendations: 1. GC-17b v0.2 should reframe T1812 honestly (the unification is the arena and methodology, not one mechanism) 2. GC-9 methodology paper should soften the “five Millennium = one duality” claim accordingly 3. T1807 (V-1 finding) should also be reframed in GC-17b v0.2

Status of GC-17b: With V-1 reframing T1807 and V-4 honesty-checking T1812, the note remains valuable but needs v0.2 to align with what’s actually proved. Lyra’s v0.2 work address both flags.

No impact on existing Millennium submissions: All seven Clay submission papers stand intact. T1812 was a meta-observation registered after the proofs, not used in them.

References

- T1812 (GC-17b registration)
 - V-1 deliverable: notes/BST_V1_Cuspidal_Test_GC17b.md
 - Paper #103 (RH proof)
 - Paper #88 (BSD proof)
 - Paper H1, H2 (Hodge proof)
 - Papers YM-A, YM-B, YM-C (YM proof)
 - BST_NS_BlowUp (NS proof)
 - BST_T1776, T1777, T1778 ($P \neq NP$ proof)
 - BST_Four_Color (Four-Color proof)
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Revision History

- v0.1 (May 13, 2026): Initial V-4 deliverable. Verdict: T1812 stays I-tier; no proof step changes; reframing recommended for GC-17b v0.2 and GC-9 methodology paper.